Disjunction behaves classically under derivability: excluded-triple inconsistency and introductions

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source's 'modus ponsens' typo is inside the span and preserved verbatim in the card; no other deviation from the printed text.

Editorial note. The label prop:provability-lor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Negation axiom not-A -> (A -> falsum). Propositional-logic part, axiom ax:lnot1. Line 967: yields the two conditional forms used to build the disjunction's refutation.
- Disjunction-elimination axiom chain. Propositional-logic part, axiom ax:lor3. Line 971: from the two refutations to (A or B) -> falsum under the pair of negated assumptions.
- Disjunction-introduction axioms. Propositional-logic part, axioms ax:lor1 and ax:lor2. Line 975 closes item (2) 'by modus ponsens' - the author's typo for 'modus ponens', kept verbatim.
- Deduction theorem. Same chapter, the deduction theorem (source line 656). Lines 968-969 and 972-973: twice used to move assumption A (respectively the triple) into an antecedent.

Source reference. Original reference ax-lnot1: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 502–502. Prerequisite pointer invoked at proof line(s) 1070; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-lor1: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 497–497. Prerequisite pointer invoked at proof line(s) 1077; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-lor2: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 498–498. Prerequisite pointer invoked at proof line(s) 1077; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-lor3: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 499–499. Prerequisite pointer invoked at proof line(s) 1074; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *1.* $!A \vee !B, \neg !A, \neg !B$ *is inconsistent.*

2. Both $!A \vdash !A \vee !B$ *and* $!B \vdash !A \vee !B$.

Proof. 1. From ax-lnot1 we get $\vdash \neg !A \to (!A \to \bot)$ and $\vdash \neg !B \to (!B \to \bot)$. So by the deduction theorem, we have $\{\neg !A\} \vdash !A \to \bot$ and $\{\neg !B\} \vdash !B \to \bot$. From ax-lor3 we get $\{\neg !A, \neg !B\} \vdash (!A \vee !B) \to \bot$. By the deduction theorem, $\{!A \vee !B, \neg !A, \neg !B\} \vdash \bot$.

2. From ax-lor1 and ax-lor2 by modus ponsens.

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